Mississippi Valley State University

# CS 341 — Discrete Structures

## Lecture 47: Guess an Exponential

*Solving Linear Recurrences*

*MCS* Section 21.3, pp. 874–883

Name: _____ Date: _____

**Where we are**

Monday produced equations and conjectures. Today produces *answers* — a procedure that derives the closed form instead of guessing it.
The Fibonacci recurrence went unsolved for almost six centuries. The method below takes about an hour to learn.

## By the end of today you should be able to

1. Write and solve the characteristic equation of a homogeneous linear recurrence.
2. Handle a repeated root.
3. Solve an inhomogeneous recurrence by the five-step method.

## 1. Homogeneous linear recurrences

**Definition**

A **homogeneous linear recurrence** has the form

$$f(n) = a_1 f(n-1) + a_2 f(n-2) + \cdots + a_d f(n-d)$$

with constants $a_i$. The number $d$ is its **order**.

**The idea.** Linear recurrences tend to have exponential solutions, so guess $f(n) = x^n$ and find which $x$ work. For Fibonacci, $f(n) = f(n-1) + f(n-2)$ gives

$$x^n = x^{n-1} + x^{n-2}.$$

Dividing by $x^{n-2}$ leaves $x^2 = x + 1$, a quadratic.

**The characteristic equation**

For $f(n) = a_1 f(n-1) + \cdots + a_d f(n-d)$, the characteristic equation is

$$x^d = a_1 x^{d-1} + a_2 x^{d-2} + \cdots + a_d.$$

Its coefficients are read straight off the recurrence.

**Theorem 21.3.1**

If $f$ and $g$ both solve a homogeneous linear recurrence, so does $sf + tg$ for any reals $s, t$.

So each root $r$ contributes a solution $r^n$, and the general solution is a linear combination of them. The boundary conditions then pin down the constants.

## 2. Worked: distinct roots

**Solve** $f(n) = 5f(n-1) - 6f(n-2)$ with $f(0) = 1$, $f(1) = 0$.

**Step 1.** Characteristic equation: $x^2 = 5x - 6$, that is $x^2 - 5x + 6 = 0$, with roots $x = 2$ and $x = 3$.

**Step 2.** General solution: $f(n) = c \cdot 2^n + d \cdot 3^n$.

**Step 3.** Apply the boundary conditions:

$$f(0) = c + d = 1, \qquad f(1) = 2c + 3d = 0.$$

Solving: $c = 3$, $d = -2$.

$$f(n) = 3 \cdot 2^n - 2 \cdot 3^n$$

**Check** $n = 2$: formula gives $12 - 18 = -6$; recurrence gives $5(0) - 6(1) = -6$. ✓

**Fibonacci, for the record**

The same method on $x^2 = x + 1$ gives roots $\frac{1 \pm \sqrt{5}}{2}$ and, after the boundary conditions, **Binet's formula**:

$$f(n) = \frac{1}{\sqrt{5}} \left( \frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left( \frac{1-\sqrt{5}}{2} \right)^n.$$

Every Fibonacci number is an integer, yet the formula is full of $\sqrt{5}$ — the irrational parts cancel exactly. The larger root $\frac{1+\sqrt{5}}{2} \approx 1.618$ is the golden ratio, and the second term shrinks to nothing, which is why consecutive Fibonacci numbers approach that ratio.

## 3. Repeated roots

**The rule**

A root $r$ of multiplicity $k$ contributes $k$ solutions:

$$r^n, \quad nr^n, \quad n^2 r^n, \ \ldots, \ n^{k-1} r^n.$$

**Solve** $f(n) = 4f(n-1) - 4f(n-2)$ with $f(0) = 1$, $f(1) = 4$.

Characteristic: $x^2 - 4x + 4 = (x-2)^2$, so $r = 2$, twice.

General solution: $f(n) = c \cdot 2^n + d \cdot n2^n$.

$f(0) = c = 1$; $\quad f(1) = 2c + 2d = 4$ gives $d = 1$.

$$f(n) = 2^n + n2^n$$

Check $n = 3$: $8 + 24 = 32$, and the recurrence gives $4(12) - 4(4) = 32$. ✓

**Why $nr^n$ and not a second $r^n$**

With a repeated root there is only *one* exponential, so $cr^n + dr^n$ collapses to a single constant times $r^n$ — not enough freedom to meet two boundary conditions. The factor of $n$ supplies the missing degree of freedom.

## 4. Inhomogeneous: the five-step method

An extra term $g(n)$ on the right makes the recurrence **inhomogeneous**: $f(n) = a_1 f(n-1) + \cdots + g(n)$.

**The procedure**

1. Drop $g(n)$ and find the roots of the characteristic equation.
2. Write the **homogeneous solution**, constants undetermined.
3. Find *any* **particular solution** to the full recurrence, ignoring the boundary conditions. Guess a form matching $g(n)$.
4. Add the two to get the **general solution**.
5. Now apply the boundary conditions to fix the constants.

**Guessing a particular solution.** If $g(n)$ is constant, try $f(n) = c$. If $g(n)$ is a polynomial, try one of the same degree. If $g(n) = 3^n$, try $c \cdot 3^n$.

**Worked (MCS).** A Hanoi variant where moving disk $k$ takes $k$ seconds: $f(1) = 1$, $f(n) = 2f(n-1) + n$.

*Steps 1–2.* Homogeneous part $f(n) = 2f(n-1)$ has characteristic $x = 2$, so the homogeneous solution is $c \cdot 2^n$.

*Step 3.* Since $g(n) = n$ is degree 1, guess $f(n) = an + b$. Substituting:

$$an + b = 2\big(a(n-1) + b\big) + n \implies 0 = (a+1)n + (b - 2a).$$

This holds for all $n$ when $a = -1$ and $b = -2$. So $-n - 2$ is a particular solution.

*Step 4.* General: $f(n) = c \cdot 2^n - n - 2$.

*Step 5.* $f(1) = 2c - 3 = 1$ gives $c = 2$.

$$f(n) = 2 \cdot 2^n - n - 2$$

**What the answer says**

The original puzzle cost $2^n - 1$. Charging by disk size costs about twice as much — not exponentially more. Making each step more expensive barely matters; it is the *number* of subproblems that drives the cost.
That observation is Friday's whole subject.

## Exit ticket

1. Write the characteristic equation of $f(n) = f(n-1) + 6f(n-2)$ and find its roots.

2. Why does a repeated root need the extra solution $nr^n$?

## Before next class

**Read** *MCS* Section 21.4, "Divide-and-Conquer Recurrences," and Section 21.5.

## Questions I still have